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Advance Physics Project Presentation

MULTI-USE SMART ALERTING SYSTEM

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B.Tech COMPUTER SCIENCE AND ENGINEERING (IOT)
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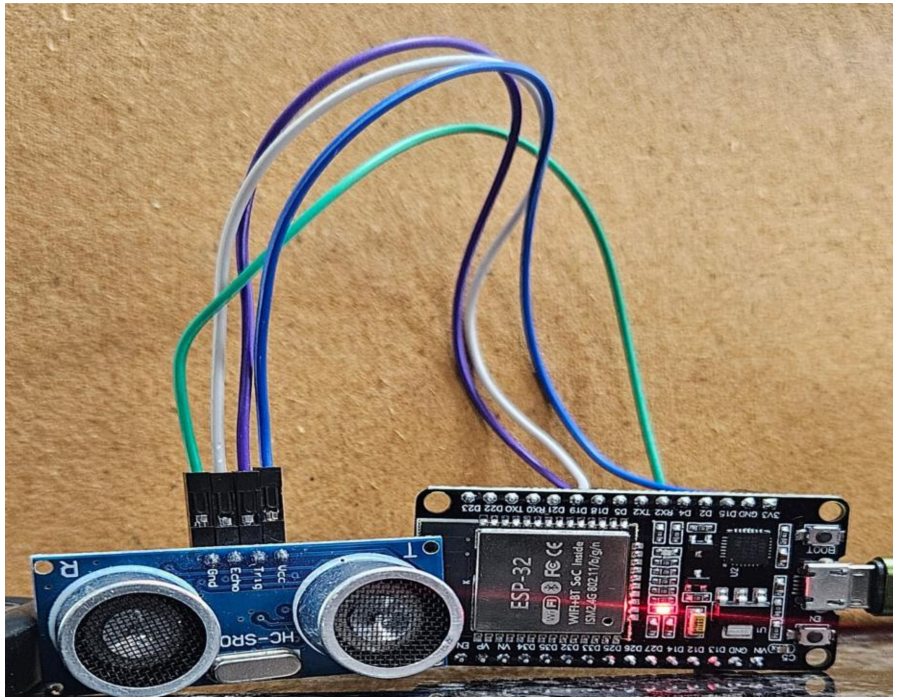

INTRODUCTION

Monitoring the level of tanks—such as water tanks, chemical containers, fuel storage, and agricultural reservoirs—is an essential requirement in homes, industries, and farms.
Traditionally, level checking is done manually, which is time-consuming and often inaccurate. With increasing demand for automation and remote monitoring, there is a need for a smarter and more efficient solution that provides real-time visibility and control.

⭐ **Advantages**

✓ One system for many detection
✓ Saves cost
✓ Real-time monitoring
✓ Can alert users anywhere via website
✓ Increases safety and reduces risk

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OBJECTIVE

To design and implement a scalable IoT-based monitoring system using ESP32, ultrasonic sensors, MQTT, and Node-RED, capable of:

1) Managing Multiple IoT Devices:

- Allowing users to **add, register, and name multiple ESP32-based sensor modules**.
- Storing device metadata (ID, name, status) for easy identification.
- Supporting future expansion without modifying the core system.

2)Real-Time Level Monitoring:

- Continuously measuring **fill level/distance** using the **HC-SR04 ultrasonic sensor**.
- Sending the level data to the MQTT broker in **real time** for processing.
- Automatically detecting and visualizing **level status** on the dashboard (e.g., *Full*, *Medium*, *Low*, *Critical*, or *Sensor Offline*)

3) Centralized Data Handling with Message Queuing Telemetry Transport(MQTT)

- Ensuring reliable, lightweight communication between devices and the server.
- Using structured MQTT topics for device telemetry and status updates.
- Supporting multiple clients simultaneously.

4) Interactive Web Dashboard with Node-RED

- Displaying real-time distance values, timestamps, and device status.
- Allowing users to **rename, edit, and manage devices dynamically**.
- Using responsive UI design so it works on
  phones, tablets, and laptops.

5) Scalable, User-Friendly IoT Architecture

- Creating a modular system that can easily add new sensors or features.
- Ensuring cross-platform accessibility through a web-based interface.
- Providing a foundation suitable for smart home, industrial, and academic IoT applications.

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Applications

Applications of Ultrasonic Sensor (HC-SR04) + ESP32

Level Monitoring Systems

- Water tank level monitoring
- Underground/overhead tank automation
- Chemical & fuel storage level
- Sump/borewell water level
- Overflow prevention

Smart Home & Automation

- Smart dustbin (auto-open + fill level)
- Parking distance assistant
- Door/room occupancy detection
- Bathroom tank monitoring

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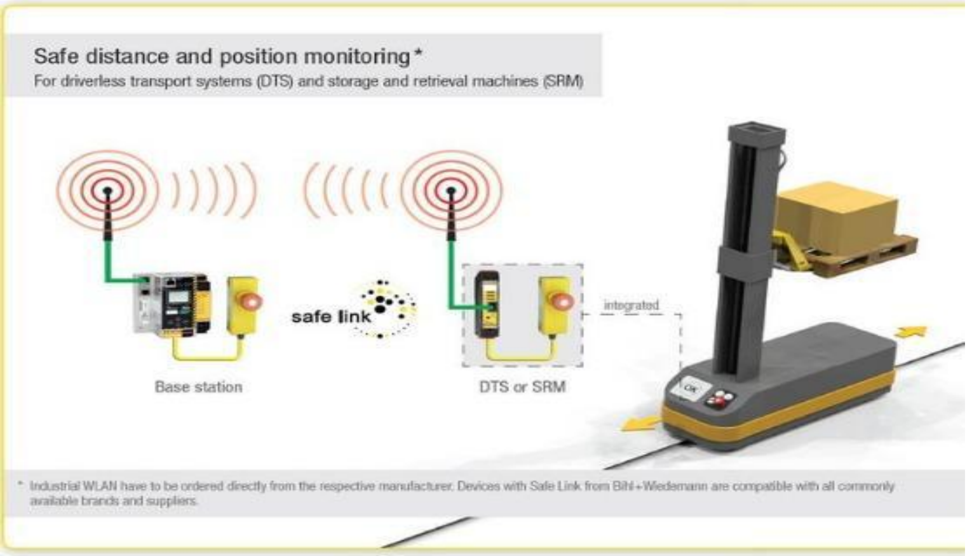
Smart Agriculture

- Irrigation tank level
- Fertilizer/pesticide solution level
- Grain/seed storage bin level

Industrial Use

- Industrial container level
- Conveyor object counting
- Robotic obstacle sensing
- Safe distance monitoring



Safety & Security

- Intruder distance detection
- Automatic gate sensing
- Door open/close alert

Smart City Solutions

- Smart parking distance systems
- Smart garbage bin level monitoring
- Flood water level detection

Hardware components used

1. ESP32 Microcontroller

- A powerful IoT development board with **built-in Wi-Fi and Bluetooth**.
- Acts as the **main controller** of the system.
- Reads data from the ultrasonic sensor.
- Sends processed data to the **MQTT broker**.
- Supports multiple sensors and low-power modes, making it ideal for continuous monitoring.

2. HC-SR04 Ultrasonic Sensor

- Measures distance by sending ultrasonic waves and receiving the ec
- Used to calculate **tank level**, **object distance**, or **fill status**.
- Provides accurate readings between **2 cm to ~800 cm**.
- Works on the principle of **time of flight** (echo return time).
- Connected to the ESP32 using trigger and echo pins.

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4. Wi-Fi Network / Router

- Required for the ESP32 to connect to the MQTT broker.
- Enables wireless communication between:
  - ESP32 → MQTT Broker
  - MQTT Broker → Node-RED Dashboard
- Allows users to access the dashboard on **mobile, laptop, or tablet** through the same network or internet.

5. Connecting Wires / Jumper Cables

- Used to connect the ultrasonic sensor to the ESP32.
  Ensures stable electrical connections for accurate readings.

6. Breadboard

Timeline of Project

- *Week 1*: General idea discussion
- *Week 2*: Research & Data Collection
- *Week 3*: Model Planning
- *Week 4*: Analysis & Final Presentation
- *Week 5* : Final presentation

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REFERENCES

- *Node Red* - Official Website: https://nodered.org
- Documentation: https://nodejs.org/api/
- NPM packages: https://www.npmjs.com
- MQTT - Official MQTT: https://mqtt.org
- ESP32 (Microcontroller) - Espressif ESP32 Docs: https://docs.espressif.com
- Coding References - W3Schools: https://www.w3schools.com

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THANK YOU!!